

Chapter 7: Special forms of square matrices Worksheet

1. Are each of the following statements true or false?

- (a) If a square matrix is both upper and lower triangular, then it must be the zero matrix.
- (b) If D is a 3×3 diagonal matrix and A is any 3×3 matrix, then $AD = DA$.
- (c) If C is a 3×3 scalar matrix and A is any 3×3 matrix, then $AC = CA$.
- (d) Every square reduced row echelon matrix is upper triangular.
- (e) Every upper triangular matrix is in reduced row echelon form.
- (f) Given a 3×3 matrix A , the matrix $A - A^T$ is symmetric.
- (g) Given a 3×3 matrix A , the matrix $A + A^T$ is symmetric.
- (h) Every skew-symmetric matrix is singular (i.e. not invertible).

2. Consider the matrix $D = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$.

- (a) Compute D^{10} .
- (b) Let $A = \begin{bmatrix} 8 & 18 \\ -3 & -7 \end{bmatrix}$. Given the information that $A = PDP^{-1}$ for $P = \begin{bmatrix} 2 & -3 \\ -1 & 1 \end{bmatrix}$, find A^{10} .

3. Recall: A 2×2 matrix $M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$ is **symmetric** if $m_{12} = m_{21}$.
- (a) Use the above definition to show the following: (i) if A, B are symmetric 2×2 matrices, then $A + B$ is a symmetric matrix, (ii) if A is a symmetric 2×2 matrix and r is any scalar, then rA is a symmetric matrix.
- (b) Show that the following statement is false by explicit counterexample: If A, B are symmetric 2×2 matrices, then AB is a symmetric matrix.
- Note: "explicit counterexample" means that we use numbers in the matrices A and B , not variables.